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The electric field $\vec{E}$ and $d\vec{A}$ point in the same direction (outward normal) at all the points on the Gaussian surface. The magnitude of $\vec{E}$ is also the same at all points due to the spherical symmetry of the charge distribution.

$$\text{Hence } E \oint_{\substack{\text{Gaussian}\\ \text{surface}}} dA = \frac{Q}{\varepsilon_0} \qquad (1.74)$$

But $\oint_{\substack{\text{Gaussian}\\ \text{surface}}} dA =$ total area of Gaussian surface $= 4\pi r^2$. Substituting this value in equation (1.74)

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0} \text{ (or) } E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

$$\text{In vector form } \vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r} \qquad (1.75)$$

The electric field is radially outward if Q > 0 and radially inward if Q < 0. From equation (1.75), we infer that the electric field at a point outside the shell will be same as if the entire charge Q is concentrated at the center of the spherical shell. (A similar result is observed in gravitation, for gravitational force due to a spherical shell with mass M)

Case (b): At a point on the surface of the spherical shell (r = R)

The electrical field at points on the spherical shell (r = R) is given by

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{r} \qquad (1.76)$$

Case (c) At a point inside the spherical shell (r < R)

Consider a point P inside the shell at a distance r from the center. A Gaussian sphere of radius r is constructed as shown in the Figure 1.42 (b). Applying Gauss law

$$\oint_{\substack{\text{Gaussian}\\ \text{surface}}} \vec{E}\cdot d\vec{A} = \frac{Q}{\varepsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0} \qquad (1.77)$$

Since Gaussian surface encloses no charge, So Q = 0. The equation (1.77) becomes

$$E = 0 \qquad (r < R) \qquad (1.78)$$

The electric field due to the uniformly charged spherical shell is zero at all points inside the shell.

A graph is plotted between the electric field and radial distance. This is shown in Figure 1.43.

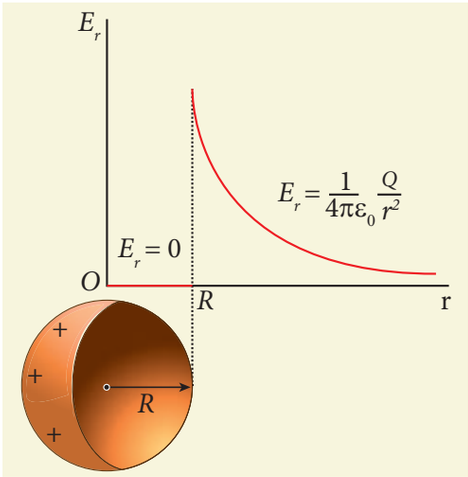


Figure 1.43 Electric field versus distance for a spherical shell of radius R

Note

Gauss law is a powerful technique whenever a given charge configuration possesses spherical, cylindrical or planer symmetry, then the electric field due to such a charge configuration can be easily found. If there is no such symmetry, the direct method (Coulomb's law and calculus) can be used. For example, it is difficult to use Gauss law to find the electric field for a dipole since it has no spherical, cylindrical or planar symmetry.

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